Fiscal Year:	FY 2012	**Task Last Updated:**	FY 10/18/2012
PI Name:	Nelson, Gregory A. Ph.D.		
Project Title:	Epigenetic Control of Radiogenic Damage Processing in C. elegans		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Loma Linda	**State:**	CA
Zip Code:	92350-1700	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 Space Radiation NNJ07ZSA001N
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/29/2012 per A. Chu/ARC and NSSC (Ed., 10/31/11) NOTE: Received NCE until 12/31/2011, per NSSC info (10/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Braby, Leslie (Texas A&M University) Ford, John (Texas A&M University)		
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Performance Goal Text:			

Task Description:	Levels of expression of specific genes in individuals are determined by the particular combination of alleles that they carry and the regulatory environment of cells expressing those genes. Molecular pathways and mechanisms that detect and respond to damage from ionizing radiation are thus subject to modulation by external processes acting upon the genes expressed in the irradiated cells. These mechanisms constrain and determine the final levels of gene activity. These epigenetic mechanisms may be cell autonomous or operate at the tissue and organism level by processing information from cytokines, growth factors, hormones, and extracellular matrix. The range of gene expression patterns driven by the combinatorial actions of these epigenetic regulatory mechanisms ultimately determines the inter-individual variability of responses to radiation exposure. We propose to address these issues using a simple animal model, the nematode C. elegans, in which genotoxicity (chromosome aberrations) can be quantified in vivo in individual intestine cells whose tissue location and cell lineage are invariant. We have shown that individual cells in this 20-cell tissue respond to genotoxic damage in their neighbors and that their radiosensitivities appear to be modulated by signals originating in an adjacent tissue, the immature gonad. To identify the mechanisms underlying these cell-cell and tissue-tissue interactions we will employ a targeted screening protocol based on RNA interference methods to determine which of approximately 250 candidate genes alter the tissue architecture, cell-cell communication properties, DNA repair capacity, and important regulatory signal cascades which we believe are likely to modulate the expression and transmission of the genotoxic damage. This will be complemented by targeted irradiations of individual cells and cell groups with microbeam and collimated alpha particles to determine the range and topology of the damage transmission between cells.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Low doses of ionizing radiation in the form of natural radioactivity, cosmic rays and medical diagnostic procedures comprise the background exposure levels for the general population. Occupational exposures may add to this dose and the sum of the two will establish health risks from radiation exposure. Risks from radiation exposure have been extrapolated from data on atomic bomb survivors, individuals exposed during radiation accidents, and older unregulated medical exposures. Regulatory limits have employed the linear no-threshold model for extrapolating risks to low doses but recent experimental findings suggest significant deviations from linearity at low doses such that the uncertainties in risk estimates must be re-evaluated. One of the important types of biological responses that may explain the deviations are those termed non-targeted effects. These are responses in which cells not directly exposed to radiation may sustain injury due to perturbations in their environment or transfer of damaging chemical species created in the irradiated cells. This task is focused directly on identifying the genes and pathways that regulate non-targeted damage formation. The high genetic homology between C. elegans and mammalian gene networks should enable predictions of homologous signaling pathways that are operational in mammals, including humans. This should, in turn, improve the understanding of the role of non-targeted effects in the low radiation dose environment and suggest mitigation strategies.
	SUPPLEMENTAL REPORTING FOR FINAL REPORT: OCTOBER 2012. Administrative: As reported previously, technical progress in the early part of the project was limited by technical difficulties in adapting the microbeam facility configuration for use with nematodes. However, in 2011 we made two important breakthroughs which enabled us to directly irradiate individual cells of the worm intestine. The first breakthrough was the construction of a transgenic worm strain that expresses green fluorescent protein (GFP) exclusively in the gut. The second was an upgrade to the optical system that allowed us to illuminate worms in the microbeam target area with blue laser light that excited the GFP-labeled cells and permitted observation with a high resolution CCD camera system. In order to take advantage of these improvements, a no-cost extension was requested to enable a multiple session experimental campaign and NASA granted this request for a new period of performance through June of 2012. We have been now able to successfully carry out three week-long microbeam irradiation sessions in which approximately 1000 animals have been targeted systematically and animals preserved and archived for analysis. Technical: In the past we have used a distance-based method to identify positions of intestinal cell pairs relative to an anatomical landmark. This year we made important improvements in methodology which enabled precise targeting of individual cells in the intestine. The major methodological improvement required a combined genetic and optical approach. The concept was to construct a transgenic worm strain that expresses green fluorescent protein only in gut cells and then visualize them. We previously reported the construction of the appropriate GFP-expressing strain called JP2003 which expresses GFP constitutively in intestine cells int-1 through int-9 under control of the gut-specific elt-2 GATA-type promoter. Two additional strains (JP2004 and JP2006) incorporating radiosensitizing mutations in genes cku-70 and smk-1 have also been constructed. With respect to optics, we tested a number of high intensity LED illumination systems and filters to the microbeam system but eventually found a blue laser with enough intensity to obliquely illuminate the small larval stage animals for visualization through the microbeam optical system and camera. Now we are able to directly visualize the desired cell(s) to target the proton beam within 1 or 2 microns. From our microbeam experiments many pair-wise combinations of cell signal transmission (determined by targeting) and reception (observed as expression of anaphase bridges) have now been observed and we have been able to construct a map of the signaling patterns between cells in the 20-cell organ. Several important observations have emerged. (1) The signaling pattern does not simply reflect which cells are in direct contact or nearest each other as distal cells may respond while proximal ones do not. (2) The distance that a signal can travel can be as far as from cell pair 2 to cell pair 8, a distance of well over 100 microns. (3) There appears to be a functional compartment boundary at approximately the position of the pharynx/intestine valve through which signals generated in pharyngeal cells to not pass or, alternatively, pharyngeal muscle and nerve cells do not generate the proper type or intensity of signal to which intestine cells react. (4) The signal transmission and reception patterns have specificity such that not every cell-cell pair interacts whether from the transmission or reception perspective. (5) The frequency of anaphase bridges in bystander cells may exceed that in directly irradiated cells - a very unexpected result. (6) The overall sensitivity pattern resembles that for whole body irradiations which may reflect the dominance of the response by bystander cells - cf. #5. Genetic screening for genes modifying the radiation-induced anaphase bridge response has identified 26 genes that modify the overall frequency of anaphase bridge expression or alter the covariant expression of bridges between cell pairs. This was established using an odds ratio-based statistical method in whole body irradiation experiments. Six of the genes exhibited strong effects. These include three DNA repair genes from the non-homologous end joining pathway for double strand breaks (cku-70, cku-80 and lig-4), a telomere structure gene (trt-1), a mitochondrial complex II gene (mev-1) and a MAP kinase interacting protein gene (smk-1). A limited number of GFP transgenic worms incorporating cku-70 and smk-1 have now been microbeam irradiated and these archived samples await further analysis.

Task Progress:	ANNUAL REPORTING PROVIDED NOVEMBER 2011: Administrative: As reported in last year's annual report, technical progress in the early part of the project was limited by technical difficulties at the Texas A&M University microbeam facility. However, in the spring of 2010 the facility was restored to normal operational status and experiments were immediately scheduled. A no-cost extension then was requested to enable a multiple session experimental campaign and NASA granted this request for a new period of performance through December 2011 and we received our final incremental funding in April. We have been able to successfully carry out three microbeam irradiation sessions to date (July 2010, October 2010, and January 2011) using different approaches to cell targeting culminating in the ability to visualized transgenic animals with fluorescent target cell nuclei. Mid-year microbeam scheduling was not practical due to staff members' competing schedules in support of NASA-funded experiments at the Brookhaven National Laboratory. To assure proper technical completion of planned experiments, a final no-cost extension was requested and NASA granted an additional 6 month extension on October 12, 2011 that extends the period of performance to June 29, 2012. This will enable us to conduct a fourth microbeam experimental run in early December 2011 and a fifth run in January/February 2012. Additional improvements in microbeam stage control and fluorescence optics have been put in place to increase experiment throughput. We have constructed new transgenic strains for use in the upcoming runs. Meanwhile the genetic screening aspects of the project proceeded at a reduced rate to conserve funds which will be sufficient to carry the project through termination. Technical: In the past we have used a distance-based method to identify positions of intestinal cell pairs. This year we made important improvements in methodology which enables precise targeting of individual cells in the intestine. The major methodological improvement required a combined genetic and optical approach. The concept was to construct a transgenic worm strain that expresses green fluorescent protein only in gut cells and then visualize them. We previously reported the construction of the appropriate GFP-expressing strain called JP2003 which expresses GFP constitutively in intestine cells int-1 through int-9 under control of the gut-specific elt-2 GATA-type promoter. With respect to optics, we tried to fit a number of high intensity LED illumination systems and filters to the microbeam system without much success but eventually found a blue laser with enough intensity to obliquely illuminate the small larval stage animals for visualization through the microbeam optical system and camera. Now we are able to directly visualize the desired cell(s) to target the proton beam. From our microbeam experiments many pair-wise combinations of cell signal transmission (determined by targeting) and reception (observed as expression of anaphase bridges) have been observed and several interesting patterns emerge. (1) The int-6 cell pair seems to transmit or receive signals at a much lower frequency than other cells in the intestine. (2) The signaling pattern does not simply reflect which cells are in direct contact or nearest each other as distal cells may respond while proximal ones do not. (3) The distance that a signal can travel can be as far as from cell pair 2 to cell pair 8, a distance of well over 100 microns. (4) There appears to be a functional compartment boundary at approximately the position of the pharynx/intestine valve. (5) The frequency of signal transmission/ reception corresponds broadly to the overall frequency of bridges observed during whole body irradiations which suggests that “in field” and “out-of-field” effects may be additive. Two more rounds of microbeam experiments are planned which will include tests of the signaling patterns for animals defective in the cku-70 and smk-1 genes which dramatically increase overall sensitivity for radiation induced anaphase bridges. Transgenic strains JP2004 and JP2006, defective in one or the other gene and expressing GFP in intestinal cells, have been constructed for this purpose. Future Plans: The next series of microbeam experiments will use three GFP-expressing strains to enable precise targeting of individual nuclei. Some experiments will test whether signal transmission and reception are affected by a DNA repair pathway or a MAP kinase signal transduction pathway. Other experiments will address properties of the signaling such as their additivity, or threshold behavior. This will continue to refine the cellular logic circuit map for signal transmission and reception that we have begun to elucidate. Meanwhile, additional genetic screening will occur to more fully reveal the genetic regulation of cell-cell signaling leading to induction of genetic damage in non-targeted cells.
Bibliography Type:	Description: (Last Updated: 03/26/2026)
Abstracts for Journals and Proceedings	Nelson GA, Jones TA, Ortloff L, Ford J, Nuñez D, Braby LA. "Microbeam studies of bystander signaling in the nematode, C. elegans." 58th Annual Radiation Research Society Annual Meeting. Rio Mar, Puerto Rico, September 30 – October 3, 2012. 58th Annual Radiation Research Society Annual Meeting. Rio Mar, Puerto Rico, September 30 – October 3, 2012. Poster 1.13. , Oct-2012
Abstracts for Journals and Proceedings	Feng S, Braby LA. "C. elegans model for bystander effects research." 58th Annual Radiation Research Society Annual Meeting. Rio Mar, Puerto Rico, September 30 – October 3, 2012. 58th Annual Radiation Research Society Annual Meeting. Rio Mar, Puerto Rico, September 30 – October 3, 2012. Poster 1.31. , Oct-2012